

Project Title: Chronic Lung Disease Predictive Analytics

Company/Organization: Teladoc Health

INTRODUCTION

About Us:

In Teladoc Health company Chronic obstructive pulmonary disease (COPD) represents a disease that can be prevented and treated, characterized by persistent airflow limitation. It has a tendency to develop progressively and is accompanied by an increased inflammatory reaction to harmful gasses like tobacco and hazardous particulates in the airways and lungs. Due to its high morbidity, disability and mortality, COPD threaten human physical and mental health. It has emerged as the fourth most deadly disease worldwide and is anticipated to turn into the third leading cause of mortality globally by 2030, causing a serious socioeconomic burden.

We have developed a Chronic Obstructive Pulmonary Disease Prediction system in Python to help users predict acute exacerbation events in the near future. Machine Learning (ML) methods were employed to construct a risk prediction model for COPD to improve its prediction efficiency. The system also integrates a chatbot that will help users know more information about the disease predicted.

The Project:

The admin can log in using their credentials. They can view all the registered users. They can add, update or delete doctors. They can view the feedback provided by the users.

The user would require to register first and then log in using their credentials. To detect COPD, the user would need to upload the sound file. As the file is uploaded, the system will detect the disease. They can view the doctors using the area-wise filter. The user can also ask for more information about the detected disease in the chatbot. They can also provide feedback to the admin.

In this system, the front end involves HTML, CSS and JavaScript and the back end involves the MSSQL database. The back-end language is Python. The framework used is Django. The dataset is used from Kaggle. The classes involved are COPD and Healthy.

The Supervised DBN Classification Machine Learning Algorithm is used to develop this system. Deep belief networks can substitute for a deep feedforward network or, given more complex arrangements, even a convolutional neural network. They have the advantages that they are less computationally expensive (they grow linearly in computational complexity with the number of layers, instead of exponentially, as with feedforward neural networks); and that they are significantly less vulnerable to the vanishing gradients problem.

Objectives:

- Predicting individuals at high risk for developing chronic lung diseases such as COPD or asthma to facilitate early intervention.
- Track disease progression in patients over time, Analyze treatment responses to personalize medication and therapy plans based on patient data
Predicting how lifestyle factors (e.g., smoking, air quality, exercise) influence disease severity and progression
- Predicting and Combine data from electronic health records, wearables, and environmental factors for a holistic view of patient health

PREREQUISITES AND REQUIREMENTS:

High-speed internet connectivity and reliable network systems are required. Secure servers for data storage and processing. Interoperable Electronic Health Records (EHR). Existing EHR systems must support data integration for seamless information flow. Mechanisms to gather real-time data from various sources (e.g., medical devices, patient wearables). Established clinical pathways for chronic lung disease management to guide analytics and interventions. Buy-in from healthcare providers, administrators, and IT staff to ensure collaboration and support.

Compliance with HIPAA and other regulations to ensure patient data privacy and security. Robust data governance policies for handling sensitive information. Algorithms and models for predictive analytics, risk stratification, and decision support. An intuitive dashboard for clinicians to easily access and interpret analytics.

Integration with monitoring devices to provide continuous patient data updates. Comprehensive training programs for staff on using the platform and interpreting data. Ongoing technical support and system maintenance to address issues and update software as needed. Implementing alert systems and recommendations based on analytics to assist clinicians in making informed decisions. Systems for collecting feedback from users to continuously improve the platform. Ensure the platform complements existing workflows without adding significant burden to healthcare providers. Tools for tracking patient outcomes and evaluating the effectiveness of interventions facilitated by the platform.

TECHNOLOGIES:

Students can independently choose suitable technologies after the design phase. However, the following technologies are suggested:

Backend: Flask, Django, Scikit-learn, TensorFlow or PyTorch
Frontend JavaScript,
Database: NoSQL, PostgreSQL or MySQL
Design: User Experience (UX) Design, User Interface (UI) Design
App Development: WebSockets or MQTT
Ops: cloud platform (e.g., AWS, Azure)

PROJECT MILESTONES:

- The MVP (minimal viable product) should be planned and implemented by the students:

- Initiation to Requirements Gathering: 1-2 months
- Technical Design to Prototype Development: 1-2 months
- Development Phases: 3-4 months
- Testing and Deployment: 1-2 months
- Post-Launch Support: Ongoing
- The exact requirements are worked out with the students based on the current requirements of the organization.